

MH2100 Calculus III

Midterm Practice 2 (Solutions)

AY 2026/2027, Semester 1

Quantitative Research Society @NTU

Practice material

Overview

- **Coverage:** curves and tangent lines; limits, continuity, partial derivatives and total differentiability; chain rule, directional derivatives, tangent lines, linear approximation and the Implicit Function Theorem; extrema and Lagrange multipliers; order reversal, polar coordinates and volume by a height double integral.
- **Practice design:** four original problems, 100 marks in total; intended for 110 minutes. This practice design is not a confirmed 2026 examination format.
- **Current administration:** Tuesday 6 October 2026, 16:30–18:20, venue TBA. One double-sided typed A4 sheet is permitted; no attachments or sticky notes are permitted. No calculator policy is stated here.

Instructions

Answer **all** questions. Show all working clearly; answers without justification may receive zero marks.

Contents

Question 1	(25 Marks)	3
Solution		3
Method 1: Direct calculation		3
UNOFFICIAL QRS marking scheme (25 marks)		3
Question 2	(25 Marks)	4
Solution		4
Method 1: Chain rule and implicit differentiation		4
UNOFFICIAL QRS marking scheme (25 marks)		4
Question 3	(25 Marks)	6
Solution		6
Method 1: Interior test and Lagrange multipliers		6
UNOFFICIAL QRS marking scheme (25 marks)		6
Question 4	(25 Marks)	7
Solution		7
Method 1: Reversal and polar height integral		7
UNOFFICIAL QRS marking scheme (25 marks)		7

Question 1**(25 Marks)****PRACTICE PROBLEM 1.**

- (a) Let $\mathbf{r}(t) = (e^t, \sin t, t^2)$. Find the point at $t = 0$, a unit tangent vector there, and an equation of the tangent line. **(8 marks)**
- (b) Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(0, 0) = 0$ and $f(x, y) = x^2y/(x^2 + y^2)$ otherwise. Determine continuity at the origin, compute both coordinate partial derivatives there, and decide total differentiability there. Justify each conclusion. **(17 marks)**

Solution**Method 1: Direct calculation**

- (a) $\mathbf{r}(0) = (1, 0, 0)$ and $\mathbf{r}'(t) = (e^t, \cos t, 2t)$, so $\mathbf{r}'(0) = (1, 1, 0)$ has length $\sqrt{2}$. Hence

$$\mathbf{T}(0) = \frac{1}{\sqrt{2}}(1, 1, 0), \quad (x, y, z) = (1, 0, 0) + s(1, 1, 0).$$

- (b) For $(x, y) \neq (0, 0)$,

$$|f(x, y)| = \frac{x^2|y|}{x^2 + y^2} \leq |y| \leq \sqrt{x^2 + y^2}.$$

Hence f is continuous at $(0, 0)$. On the axes,

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0)}{h} = 0, \quad f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k)}{k} = 0.$$

A total derivative would therefore be zero. Along (t, t) ,

$$\frac{|f(t, t)|}{\sqrt{2}t^2} = \frac{1}{2\sqrt{2}},$$

which does not tend to zero. Thus

$$f \text{ is continuous but not totally differentiable at } (0, 0).$$

UNOFFICIAL QRS marking scheme (25 marks)

- (a) **(8 marks)**
- (i) **(2 marks)** point;
 - (ii) **(3 marks)** tangent vector and normalization;
 - (iii) **(3 marks)** tangent line.
- (b) **(17 marks)**
- (i) **(5 marks)** radial bound proving continuity;
 - (ii) **(4 marks)** partial derivatives;
 - (iii) **(5 marks)** differentiability remainder;
 - (iv) **(3 marks)** conclusions.

Question 2**(25 Marks)****PRACTICE PROBLEM 2.**

- (a) Let h be C^2 near $(0, 1)$, with $h(0, 1) = 2$, $\nabla h(0, 1) = (3, 2)$, and $h_{uu}(0, 1) = 2$, $h_{uv}(0, 1) = 1$, $h_{vv}(0, 1) = -3$. Define $G(x, y) = h(x - y, xy)$. At $P = (1, 1)$, find $G_{xy}(P)$, the directional derivative in the unit direction from P to $(3, 0)$, the tangent line to $G = 2$, and the linear approximation of $G(1.01, 0.98)$. **(15 marks)**
- (b) Let $F(x, y, z) = x + y^2 + e^z - 2$. Use the IFT near $(0, 1, 0)$ to write $z = q(x, y)$. Find $q_x(0, 1)$ and $q_y(0, 1)$, then linearly approximate $q(0.03, 0.99)$. **(10 marks)**

Solution**Method 1: Chain rule and implicit differentiation**

- (a) At P , $(x - y, xy) = (0, 1)$. Since $(u_x, u_y) = (1, -1)$ and $(v_x, v_y) = (1, 1)$,

$$\nabla G(P) = (3 + 2, -3 + 2) = (5, -1).$$

The required unit direction is $(2, -1)/\sqrt{5}$, giving

$$D_{\mathbf{u}}G(P) = \frac{(5, -1) \cdot (2, -1)}{\sqrt{5}} = \frac{11}{\sqrt{5}}.$$

The tangent line is $\boxed{5(x - 1) - (y - 1) = 0}$. Since $L(x, y) = 2 + 5(x - 1) - (y - 1)$,

$$\boxed{G(1.01, 0.98) \approx L(1.01, 0.98) = 2.07}.$$

For the mixed derivative, differentiate $G_x = h_u + yh_v$ with respect to y :

$$G_{xy} = h_{uu}u_y + h_{uv}v_y + y(h_{vu}u_y + h_{vv}v_y) + h_v.$$

At P , $u_y = -1$, $v_y = 1$, $y = 1$, so

$$\boxed{G_{xy}(P) = 2(-1) + 1(1) + (1(-1) + (-3)(1)) + 2 = -3}.$$

- (b) $F(0, 1, 0) = 0$ and $F_z(0, 1, 0) = 1 \neq 0$, so the IFT applies. Implicit differentiation gives

$$q_x = -\frac{F_x}{F_z} = -1, \quad q_y = -\frac{F_y}{F_z} = -2 \quad \text{at } (0, 1).$$

Thus

$$\boxed{q(0.03, 0.99) \approx 0 - 0.03 - 2(-0.01) = -0.01}.$$

UNOFFICIAL QRS marking scheme (25 marks)

- (a) **(15 marks)**
- (i) **(4 marks)** chain-rule gradient;
 - (ii) **(3 marks)** unit direction and derivative;
 - (iii) **(2 marks)** tangent line;
 - (iv) **(2 marks)** approximation;

- (v) **(4 marks)** mixed second derivative.
- (b) **(10 marks)**
 - (i) **(3 marks)** IFT hypotheses;
 - (ii) **(4 marks)** derivatives;
 - (iii) **(3 marks)** approximation.

Unofficial – QRS@NTU – qrsntu.org

Question 3**(25 Marks)****PRACTICE PROBLEM 3.** Let $f(x, y) = x^2 - y^2$ on $C = \{(x, y) \in \mathbb{R}^2 : x^2 + 9y^2 \leq 9\}$.

- (a) Find and classify every interior critical point. **(8 marks)**
- (b) Use Lagrange multipliers to find every extremum candidate on $x^2 + 9y^2 = 9$. **(12 marks)**
- (c) Use the Extreme Value Theorem to state all absolute extrema on C . **(5 marks)**

Solution**Method 1: Interior test and Lagrange multipliers**

- (a) $\nabla f = (2x, -2y)$ vanishes only at $(0, 0)$. Here $D = f_{xx}f_{yy} - f_{xy}^2 = 2(-2) - 0 = -4 < 0$, so $(0, 0)$ is a saddle point.
- (b) The boundary $x^2 + 9y^2 = 9$ is a C^1 regular constraint because its gradient $\langle 2x, 18y \rangle$ is nonzero there. The equations are

$$2x = 2\lambda x, \quad -2y = 18\lambda y, \quad x^2 + 9y^2 = 9.$$

If $y = 0$, then $x = \pm 3$ and $f = 9$; if $x = 0$, then $y = \pm 1$ and $f = -1$. The simultaneous nonzero case would require $\lambda = 1$ and $\lambda = -1/9$, impossible. Thus the only candidates are

$$(\pm 3, 0) \text{ with } f = 9, \quad (0, \pm 1) \text{ with } f = -1.$$

- (c) f is continuous and C is compact, so EVT applies. Comparing with $f(0, 0) = 0$,

$$\max_C f = 9 \text{ at } (\pm 3, 0), \quad \min_C f = -1 \text{ at } (0, \pm 1).$$

UNOFFICIAL QRS marking scheme (25 marks)

- (a) **(8 marks)**
- (i) **(3 marks)** critical point;
 - (ii) **(3 marks)** discriminant;
 - (iii) **(2 marks)** classification.
- (b) **(12 marks)**
- (i) **(4 marks)** Lagrange system;
 - (ii) **(4 marks)** all branches;
 - (iii) **(4 marks)** candidates and values.
- (c) **(5 marks)**
- (i) **(2 marks)** EVT hypotheses;
 - (ii) **(3 marks)** global extrema.

Question 4**(25 Marks)****PRACTICE PROBLEM 4.**

- (a) Reverse the order of integration and evaluate

$$\int_0^1 \left[\int_{\sqrt{y}}^1 (1+x) dx \right] dy.$$

(10 marks)

- (b) Let
- D
- be the part of
- $x^2 + y^2 \leq 4y$
- in the first quadrant. Find the volume above
- D
- and below
- $z = 20 - x^2 - y^2$
- , using polar coordinates and a height double integral.
- (15 marks)**

Solution**Method 1: Reversal and polar height integral**

- (a) The region is
- $0 \leq y \leq 1$
- ,
- $\sqrt{y} \leq x \leq 1$
- , equivalently
- $0 \leq x \leq 1$
- ,
- $0 \leq y \leq x^2$
- . Therefore

$$\int_0^1 \int_{\sqrt{y}}^1 (1+x) dx dy = \int_0^1 \int_0^{x^2} (1+x) dy dx = \boxed{\int_0^1 (x^2 + x^3) dx = \frac{7}{12}}.$$

- (b) The circle is
- $r^2 = 4r \sin \theta$
- . In the first quadrant,

$$0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq r \leq 4 \sin \theta.$$

The height is positive because $r \leq 4$. Thus

$$V = \int_0^{\pi/2} \int_0^{4 \sin \theta} (20 - r^2) r dr d\theta = \int_0^{\pi/2} (160 \sin^2 \theta - 64 \sin^4 \theta) d\theta = \boxed{28\pi}.$$

UNOFFICIAL QRS marking scheme (25 marks)

- (a)
- (10 marks)**

- (i) **(3 marks)** reversed region;
- (ii) **(3 marks)** bounds;
- (iii) **(2 marks)** inner integral;
- (iv) **(2 marks)** value $7/12$.

- (b)
- (15 marks)**

- (i) **(4 marks)** polar geometric reasoning;
- (ii) **(3 marks)** height and Jacobian;
- (iii) **(5 marks)** double integral;
- (iv) **(3 marks)** value 28π . Equivalent correct Cartesian work receives full credit.